Discrete Structure

Unit 2: Logical Reasoning and Proof Techniques

2.2 Proof Methods

A proof is a method for establishing the truth of a statement. It is a logical explanation that shows **why a mathematical statement is always true**, not just for some values but for **all possible cases**.

Basic Terminology

Here are the Basic Terminologies in Proof (in short):

- Statement / Proposition: A sentence that is either true or false.
- Truth Value: Indicates whether a statement is true (T) or false (F).
- Assumption / Hypothesis: What is taken as true at the start of a proof.
- Conclusion: What needs to be proved.
- Implication: "If P, then Q" ($P \Rightarrow Q$).
- Converse: "If Q, then P" ($Q \Rightarrow P$).
- Contrapositive: "If not Q, then not P" ($\neg Q \Rightarrow \neg P$).
- Contradiction: A statement that is always false.
- Lemma: A small result used to prove a larger theorem.
- Theorem: A statement that has been proved true.
- Corollary: A result that follows directly from a theorem.

Proof Methods :

1. Direct Proof
2. Proof by Contrapositive
3. Proof by Contradiction
4. Proof by Mathematical Induction
5. Proof by Cases

1. Direct Proof

A **direct proof** is a method in which the given hypothesis is assumed to be true and, by using definitions, axioms, and known results, the conclusion is obtained logically.

Logical Form:

If $P \Rightarrow Q$, assume $P$and prove Q.

Steps:

1. Assume the hypothesis.
2. Use definitions and known results.
3. Arrive at the conclusion.

Example:

Statement: If $n$is an even integer, then $n^2$is even.

Proof:

Assume n is even. Then $n = 2k$ for some integer k.

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Hence, $n^2$is even.

2. Indirect Proof

An **indirect proof** is a method in which a statement is proved by proving an equivalent statement or by assuming the opposite and reaching a contradiction.

a) Proof by Contraposition

To prove $P \Rightarrow Q$, we prove its contrapositive $\neg Q \Rightarrow \neg P$.

A statement and its contrapositive are logically equivalent.

Steps:

1. Assume $\neg Q$.
2. Show that $\neg P$ follows.
3. Conclude that $P \Rightarrow Q$ is true.

Example:

Statement: If $n^2$is even, then n is even.

Proof:

Contrapositive: If $n$is odd, then $n^2$is odd.

Assume $n = 2k + 1$for some integer k.

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$$

This is odd. Hence, the original statement is true. ■

b) Proof by Contradiction

In a **proof by contradiction**, we assume that the given statement is false and show that this leads to a contradiction. Therefore, the original statement must be true.

Steps:

1. Assume the statement is false.
2. Deduce a contradiction.
3. Conclude the statement is true.

Example 1: Prove by Contradiction, If n is an integer and $3n + 2$is odd, then $n$is odd.

Let n be an integer.

Assume that

$$3n + 2 \text{ is odd}$$

but **suppose, for contradiction**, that

$$n \text{ is even.}$$

Since n is even, there exists an integer k such that

$$n = 2k$$

Substitute this into $3n + 2$:

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$$

This expression is divisible by 2, so:

$$3n + 2 \text{ is even}$$

But this **contradicts** our assumption that $3n + 2$ is odd.

Conclusion

Since assuming $n$is even leads to a contradiction, our assumption must be false.

Therefore,

$$\boxed{\text{If } 3n + 2 \text{ is odd, then } n \text{ is odd.}}$$

Example 2:
Statement: $\sqrt{2}$is irrational.

Proof:
Assume $\sqrt{2}$is rational. Then

$$\sqrt{2} = \frac{a}{b}$$

where a and b are coprime integers.

$$2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$$

So $a$is even. Let $a = 2k$.
Then $b^2 = 2k^2$, so $b$is also even.
This contradicts the assumption that $a$and $b$are coprime.
Hence, $\sqrt{2}$is irrational. ■

3) Proof by mathematical induction

Mathematical induction is a method used to prove that a statement $P(n)$is true for all natural numbers

n = 1,2,3….

Steps of Mathematical Induction

To prove a statement $P(n)$is true for all $n \geq k$:

Step 1: Base Case

Verify that the statement is true for $n = n_0$.

$$P(n_0) \text{ is true.}$$

Step 2: Induction Hypothesis

Assume that the statement is true for $n = k$, where $k \geq n_0$.

$$P(k) \text{ is true.}$$

(This assumption is called the **induction hypothesis**.)

Step 3: Induction Step

Using the induction hypothesis, prove that the statement is true for $n = k + 1$.

$$P(k) \Rightarrow P(k+1)$$

Step 4: Conclusion

Since the statement is true for $n = n_0$and true for $k + 1$whenever it is true for k, the statement is true for all $n \geq n_0$.

$$\therefore P(n) \text{ is true for all } n \in \mathbb{N}.$$

Example 1

Prove by mathematical induction that:

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Solution

Let

$$P(n): 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Step 1: Base Case ($\boldsymbol{n = 1}$)

$$\text{LHS} = 1$$
$$\text{RHS} = \frac{1(1+1)}{2} = 1$$
$$\therefore P(1) \text{ is true.}$$

Step 2: Induction Hypothesis

Assume that:

$$P(k): 1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

Step 3: Induction Step

Consider $P(k+1): 1 + 2 + \cdots + k + (k+1)$

Using the hypothesis:

$$= \frac{k(k+1)}{2} + (k+1)$$
$$= \frac{(k+1)(k+2)}{2}$$
$$= \frac{(k+1)((k+1)+1)}{2}$$
$\therefore P(k+1)$ is true.

Step 4: Conclusion

Hence, by mathematical induction,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Example 2

Prove that

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

Let

$$P(n): 1 + 3 + 5 + \cdots + (2n-1) = n^2$$

Base Case:
For $n = 1$,

$$1 = 1^2$$

So, $P(1)$is true.

Induction Hypothesis:
Assume $P(k)$is true, i.e.,

$$1 + 3 + \cdots + (2k - 1) = k^2$$

Induction Step:

$$\begin{aligned} &1 + 3 + \cdots + (2k - 1) + (2k + 1) \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

Thus, $P(k + 1)$is true.

Conclusion:
Hence, the result holds for all $n \in \mathbb{N}$.

Example 3:
Show that

$$1^2 + 3^2 + 5^2 + \cdots + (2n + 1)^2 = \frac{(n + 1)(2n + 1)(2n + 3)}{3}$$

using mathematical induction.

Let

$$P(n): 1^2 + 3^2 + 5^2 + \cdots + (2n + 1)^2 = \frac{(n + 1)(2n + 1)(2n + 3)}{3}$$

Base Case

For $n = 0$:

$$\begin{aligned} &\text{LHS} = (2 \cdot 0 + 1)^2 = 1^2 = 1 \\ &\text{RHS} = \frac{(0 + 1)(2 \cdot 0 + 1)(2 \cdot 0 + 3)}{3} = \frac{1 \cdot 1 \cdot 3}{3} = 1 \end{aligned}$$

Hence, $P(0)$is true.

Induction Hypothesis

Assume $P(k)$is true for some $k \geq 0$, i.e.,

$$1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$$

Induction Step

Consider $P(k+1)$:

$$1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2 + [2(k+1)+1]^2$$

Using the hypothesis:

$$= \frac{(k+1)(2k+1)(2k+3)}{3} + (2k+3)^2$$

Factor $(2k+3)$:

$$= \frac{(2k+3)((k+1)(2k+1) + 3(2k+3))}{3}$$

Simplify inside the bracket:

$$(k+1)(2k+1) + 3(2k+3) = 2k^2 + k + 2k + 1 + 6k + 9 = 2k^2 + 9k + 10$$

Also, notice: $(k+2)(2k+3)(2k+5)/3$is the formula for $P(k+1)$

Check LHS simplifies to:

$$\frac{(k+2)(2k+3)(2k+5)}{3}$$

✓ Hence $P(k+1)$is true.

Conclusion

By mathematical induction,

$$1^2 + 3^2 + 5^2 + \cdots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$$

for all $n \geq 0$.